Francesco Bordoni: Technical Contribution & Impact

Audi Hungaria

Audi Development Camp 2026 – Team 03

⚙️ Technical Tasks & Responsibilities

- **Baseline Engineering:** Authored the deterministic Python scripts for Level 02 (Standing Drop) and Level 03 (2-Step Jump Throw) using MuJoCo.
- **Data & Analytics:** Defined success criteria metrics, programmed telemetry extraction and generated Python evaluation plots to compare baseline vs. RL performance.
- **Validation:** Executed physics validations, documented constraints, and managed the Trello workflow to ensure seamless integration with the RL team.

🛠️ Problems Solved

- **Dynamic Stability:** Resolved "falling due to jump" and walking weight-transfer issues by implementing a PD Gyroscope anti-drift system and linear keyframe interpolation.
- **Simulation Fidelity:** Fixed clipping and invalid floor-crossing by enabling physics micro-stepping and mapping all 29 hardware motors.
- **Team Alignment:** Acted as the technical bridge, continuously translating physics constraints to the AI team to unblock their training pipelines.

🧠 Lessons Learned

- **Technical:** Gained applied knowledge of Reinforcement Learning mechanics and the hidden complexities of Sim2Real environment validation.
- **Soft Skills:** Refined executive technical presentation skills and the ability to justify engineering trade-offs to senior industry judges.

💻 Tools & Technologies

- **Core:** Python, MuJoCo Physics Engine, VS Code, Git / GitHub.
- **Management & Docs:** Trello (Agile tasks), PowerPoint, Word.
- **AI Support:** Free version of Google AI Studio for rapid syntax comprehension and algorithm debugging.

🏆 Key Achievements

- Successfully coded highly robust, self-balancing humanoid baselines ahead of schedule.
- Maintained strict adherence to project timelines, proving that structured task management guarantees high-quality deliverables under severe time pressure.